

The Kodamai Container Program: A Complete Technical Record

Part VI — Open Questions & Frontier

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2026-10-10

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Abstract

Part VI records the honest edge of the program: what is conjectural, what is refuted, and what the next moves are. A refutation is a **[proved]** *negative*, not an open question; standing open conjectures are distinguished from results actively refuted. Grades are copied verbatim.

Thesis of Part VI. The honest edge of the program. Distinguish standing open conjectures from results we have actively refuted (a refutation is a **[proved]** *negative*, not an open question).

1 The two standing open registry fronts

Result 1.1 (Equivalence chain M5/M6, **[in-progress]**). Node **equivalence-chain**. M5 (bridge to Poly / four monoidal structures, = Clarke–Di Meglio) and M6 (double category Org, dynamic categories) are OPEN. The internal replacement theorem (“internal directed container $\simeq$ internal category $\simeq$ comonad in C ” for general symmetric monoidal C) is conjectural (SEED open Q1).

Result 1.2 (Infinite-arity closed-tensor classification, **[in-progress]**). Node **closed-tensor-classification**. Bounded arity is settled ($\times$ or $\vee_S$); the infinite-arity boundary is an honest open problem.

2 SEED open questions (Path-level)

- **Distributive-law decision procedure:** for which pairs of directed containers does a distributive law exist? ZS (L)+(G) give checkable criteria; the general theory is incomplete (SEED Q2; *pairwise-zs* is the current best). *OPEN*.
- **q -calibration:** what is q (braided structure) for real concurrent systems? Empirical, *OPEN* (SEED Q3).
- **Density comonads = polynomial comonads** (Spivak 2025 conjecture) — ORTHOGONAL to the seed per connection **density-comonads-orthogonal-seed-q5**; *OPEN* (SEED Q5).
- **Coinductive polynomial trees** (Spivak 2026) vs. adaptive GA topologies — *OPEN* (SEED Q6).

3 Refuted / corrected (negative results — state as such, not as open)

Result 3.1 (Crown Stage 2, **[proved]** (negative)). Branch-shape Ψ FAILS (over-counts by $2^{\Sigma(\lambda-1)}$); the one-shape Ψ' is positive over $\mathcal{T}$; $k \geq 2$ separate — sharing = descent, not coproduct of branches. State the refutation, not the original crown. Connection `crown-stage2-unfolding-vs-descent` (shipped note).

Result 3.2 (Lanfranchi parallelizability port, **[proved]** (negative)). The Lanfranchi parallelizability port is ILL-POSED (node `lie-parallelizable-poly-comonoid`; see Part III).

Result 3.3 (Crown TFAE, **[proved]** (negative)). Crown TFAE is FALSE $\rightarrow$ a strict 4-level chain (MEMORY `crown-tfae-strict-chain`).

Result 3.4 (The $[\omega]$ -as-non-associativity demotion, 2026-10-09). Carried in Part II; the corrected reading is *splitting*, not non-associativity.

4 Independence-flavoured / speculative frontier

- **Conjecture 6.2 / Gap-S** (is $F = \text{Hom}(E, \oplus_{\mathbb{N}} -)$ projective?): reformulated $\text{Ext}^1 \rightarrow \text{Hom}$, one non-liftable φ suffices in ZFC; Gap-S is FALSE under CH; current bet $\mathcal{E} = 0$ in ZFC via an automorphism-rigidity analogy — *OPEN crux* (`vec-admissibility-not-zfc.tex`, MEMORY `gapS-false-under-CH`).
- **Clio's peer-claimed results** (transfer operators $\leftrightarrow$ container derivatives) **[peer-claimed]** (node `clio-peer-claims`) — register at peer-claimed; do NOT promote without an independent check.